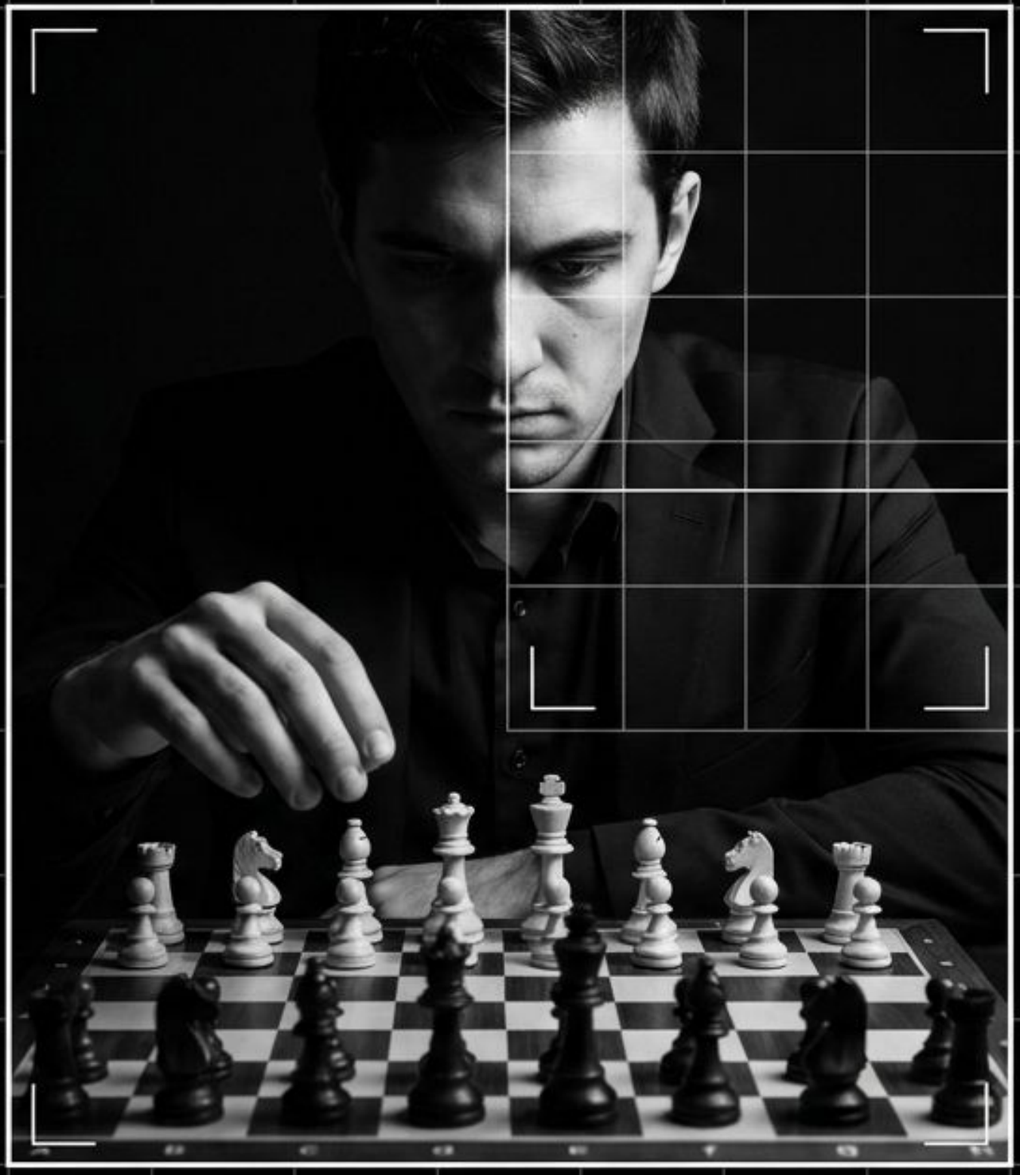
Decoding the Human Engine

AI-Driven Cognitive Load Detection in Elite Chess

Elizaveta Khoroshilova | Project Lead | Applied AI Solutions Development



Agenda

- Problem Statement
- Why this problem
- Literature Review & Past Projects
- ML Canvas
- Model Benchmarking
- Model Deployment
- Challenges & Issues Encountered
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Problem Statement: The Engine Blindspot

The Engine

Modern engines like Stockfish evaluate the mathematical position of the board flawlessly, but possess zero visibility into the human playing the game.

The Human Variable

Under extreme cognitive load and time pressure (Zeitnot), players experience severe physical and psychological fatigue that precedes critical mistakes.

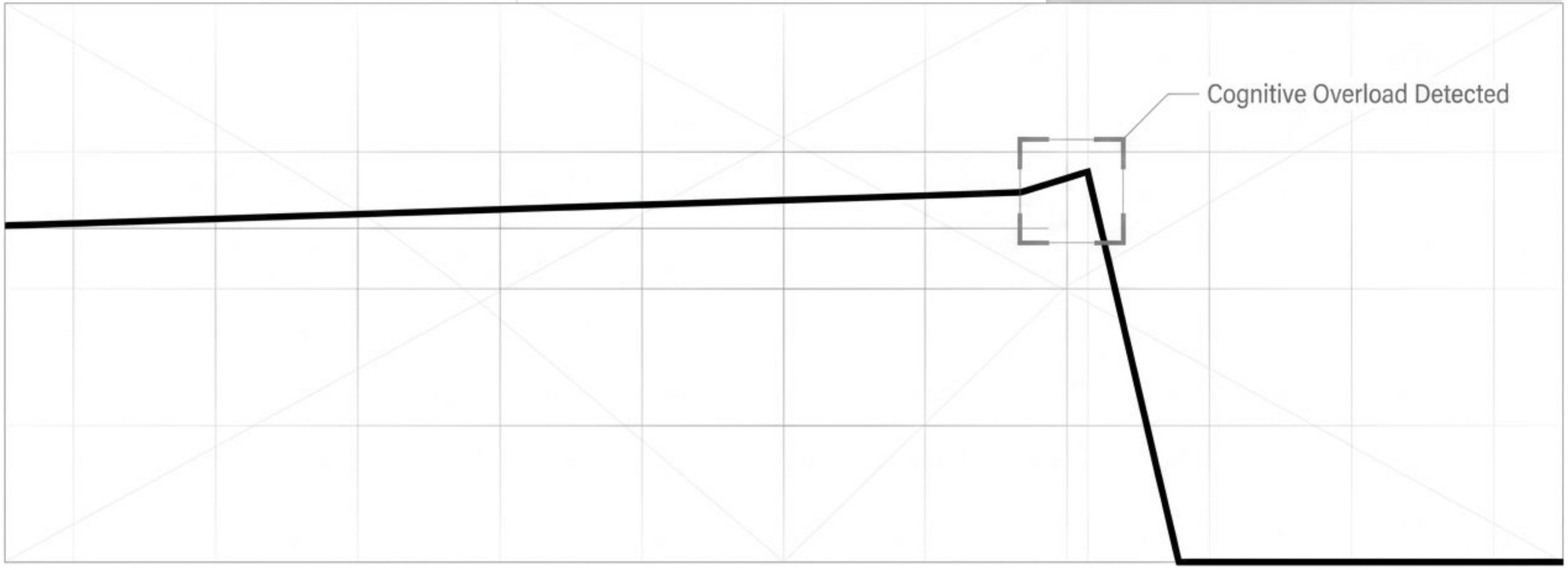
The Data Gap: Broadcasts and betting platforms currently rely entirely on reactive board analytics, possessing zero biometric data to measure human cognitive strain.

Why This Problem: Predicting the Unpredictable

The Business Case: Live esports broadcasting and chess betting markets lose predictability and revenue when elite players unexpectedly blunder.

The Shift (Reactive to Predictive): Current analytics react to a blunder after a piece is moved. We aim to predict the mistake before the piece is even touched.

Value Proposition: By quantifying the physical manifestations of stress, we generate a real-time Blunder Probability Index (BPI), creating a massive competitive advantage for sportsbooks and coaches.



Literature Review & Past Projects: Defining Focus

Past Limitations: Previous cognitive load studies rely on invasive wearables (ECG, EEG) which are strictly banned in professional chess. **Our innovation is 100% non-invasive computer vision based on Kahneman's Framework.**

System 1 (Normal Focus)

Intuitive play, recording moves. Marked by relaxed posture, vertical spinal alignment, and high physical variance.

System 2 (Deep Focus)

Active calculation and high mental demand. Marked by extreme forward lean (visual narrowing), hand-to-face anchoring (relieving neck fatigue), and strict postural rigidity.

ML Canvas: Quality Over Quantity

Data Curation

- Sourced 1,500+ raw frames from FIDE live broadcasts.
- Curated down to a 230-frame **Golden Dataset** to explicitly eliminate noisy, transitional states and focus entirely on absolute **System 1 vs. System 2** postures.

Custom Augmentations

- **CLAHE & Brightness**: Ensuring robustness across varying tournament hall lighting conditions.
- **Gaussian Blur & Flip**: Simulating broadcast compression and varied camera angles.
- **Cutout**: Randomly blacking out image sections to simulate occlusion (hands covering faces), forcing the model to learn broader postural geometry.

Architecture

YOLO11n (Nano) chosen specifically to prevent prevent overfitting on a small dataset and to guarantee ultra-fast real-time inference.

Model Benchmarking:
Metric Validation

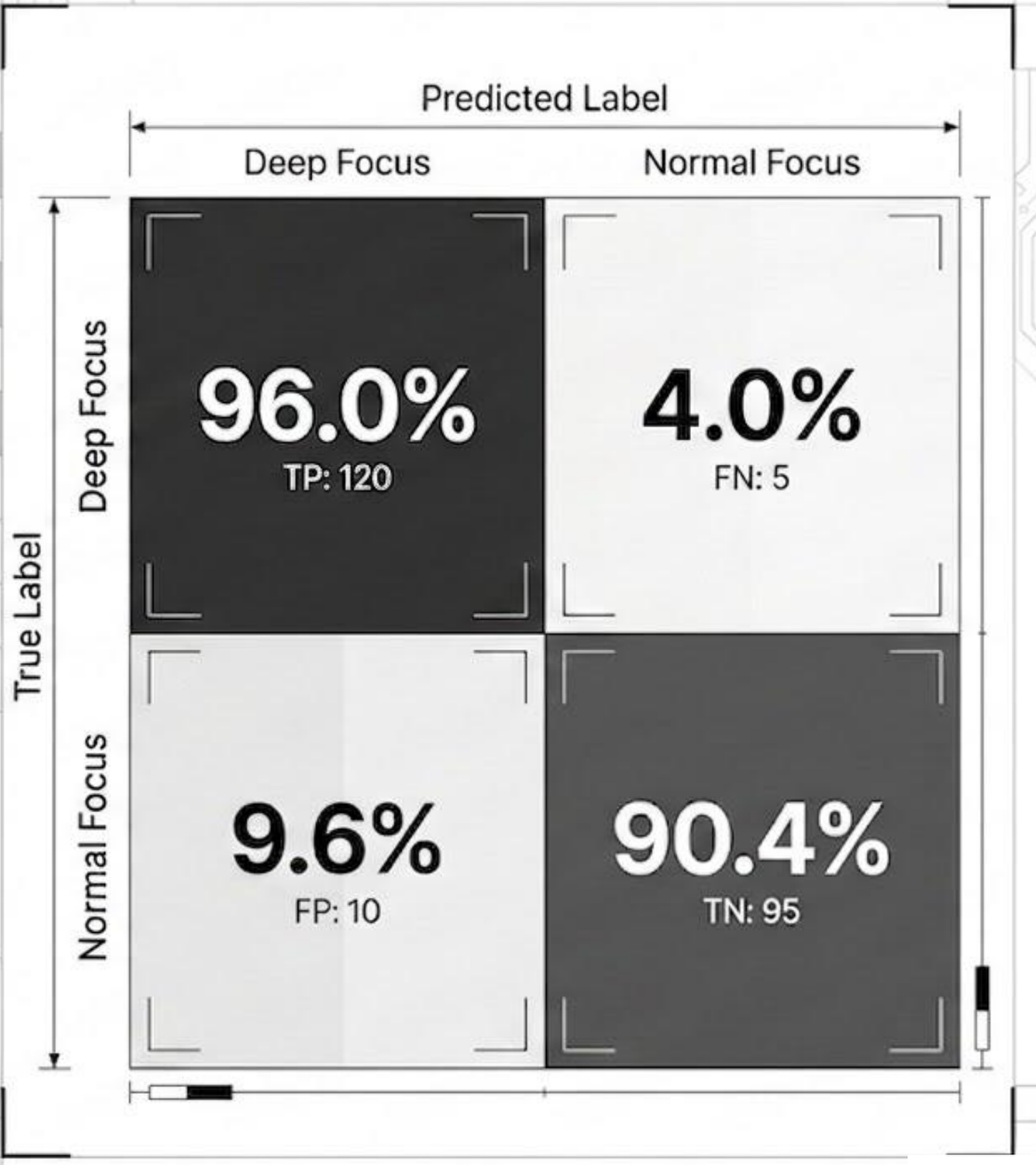
Latest Test Set Metrics:

mAP@50:	Precision:
92.3%	85.4%
Recall:	**F1 Score:**
96.0%	90.4%

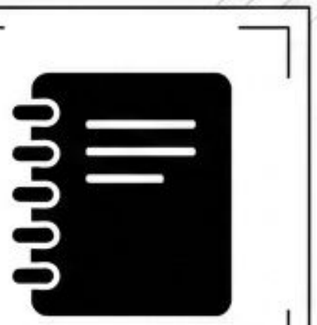
Data Science Insight:

The model is exceptionally precise at detecting Deep Focus. High cognitive stress forces the human body into rigid, predictable geometric postures. Conversely, the relaxation of Normal Focus acts as a strong baseline, proving our hypothesis that physical rigidity is a highly trackable biomarker.

Model: YOLOv11n (Nano) Parameters: 2.6M Inference Speed: ~10ms



Model Deployment Pipeline



1. Notebook

Rapid prototyping, data engineering, and initial YOLO11 weight generation.

2. Script

Extraction of YOLO inference logic into modular, production-ready Python files.

3. Flask

Wrapping the inference engine in a RESTful API to accept multipart/form-data image payloads.

4. Docker

Containerizing the Flask application and system dependencies for absolute environmental reproducibility.

5. Host in Cloud

Deployment of the containerized image to a serverless environment (Google Cloud Run / Render) for scalable, global API accessibility.

Challenges & Issues Encountered

1. Class Imbalance

Issue:
Broadcasts show “Normal Focus” 80% of the time, risking majority-class bias.

Solution:
Addressed via strategic manual labeling and Cutout augmentations to synthetically expand the minority class.

2. API Payload Limits

Issue:
Sending full 2-minute video files crashed the synchronous REST API (**Request Entity Too Large**).

Solution:
Implemented Keyframe Extraction to send only lightweight, optimal frames (Phase 2 roadmap includes local Edge deployment).

3. False Positives

Issue:
Transitional movements (briefly adjusting glasses) were flagged as deep focus.

Solution:
Tuned the Confidence Threshold during inference to prioritize Precision over Recall, eliminating noise.

Demo: DeepFocus API in Action

Q&A

Thank you. Ready for questions.

Project Repository: github.com/l-horosh/chess_candidates